

An analytics framework for interpretable subseasonal forecasting under decadal climate variability

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ABSTRACT

Subseasonal forecasts (SSFs) are essential for managing climate risk. However, their precision is influenced by decadal climatic variability and long-term climate change. As climate change continues, this challenge is expected to grow. To address this, we apply explainable artificial intelligence (XAI) to improve transparency and understanding of deep learning models in climate forecasting. XAI helps reveal how models make predictions, supporting interpretability and scientific reliability. This study uses a Convolutional Long Short-Term Memory 2D (ConvLSTM2D) deep learning model to predict European SSFs and compares it to a baseline Random Forest model. ConvLSTM2D is a deep learning architecture that integrates Convolutional Neural Networks (CNNs) for capturing spatial patterns with Long Short-Term Memory (LSTM) networks for modelling temporal dependencies. Both models are trained on high-resolution European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5 (ERA5) data. We apply Gradient-weighted Class Activation Mapping (Grad-CAM) and Feature Importance Analysis (FIA) to explain the model decision-making process and identify key meteorological drivers. For 28-day lead 2 m temperature anomalies, the deterministic skill of the ConvLSTM2D model is modest ($R^2 = -0.0424$) and lower than that of the Random Forest baseline ($R^2 = 0.5624$), highlighting the strength of relatively simple tree-based methods in this data- and noise-limited subseasonal prediction setting. Nevertheless, XAI provides useful insights by identifying meteorologically relevant patterns, such as synoptic systems and moisture gradients. FIA highlights important variables such as sea level pressure, 2 m temperature, and total column water vapour. Temporal feature importance also shows the model's reliance on historical data, with both a recency effect and a persistent influence from past time steps. These findings show that XAI can provide valuable diagnostic information for deep learning models in climate science, helping to refine model design and offering a path towards more trustworthy and transparent SSF systems for Europe and elsewhere in a changing climate.

1. Introduction

Accurate and reliable weather and climate forecasts are essential for modern society. They support decision-making across sectors such as energy, agriculture, public health, and disaster preparedness. In Europe, where climate conditions vary significantly and extreme weather events pose growing risks, the demand for high-quality forecasts is especially acute [1,2].

While traditional weather forecasts typically focus on short-term timescales and seasonal predictions offer long-range insights, a crucial forecasting gap persists in the *subseasonal range*, spanning approximately two weeks to two months [1]. Subseasonal forecasts (SSFs) are increasingly recognized for their value in bridging this temporal gap, providing lead time that is vital for proactive climate risk management [2,3]. The importance of SSFs continues to rise with the growing influence of climate change, which has intensified the

frequency and severity of extreme weather across Europe. Southern Europe, for example, has seen more persistent droughts, while northern regions face increased flooding. These shifts underscore the urgency of improving SSFs to support adaptation strategies [4,5].

Enhancing SSF accuracy, however, remains challenging due to the inherent complexity of atmospheric processes and their nonlinear interactions [1,3]. A key factor influencing SSF performance is *decadal climate variability*, natural shifts in climate patterns that occur over timescales of 10 to 30 years [6,7]. Large-scale oscillations, such as the Atlantic Multidecadal Oscillation (AMO) and the North Atlantic Oscillation (NAO), have well-established effects on regional weather patterns and are known to influence both seasonal and subseasonal forecasts [6–8]. The NAO, for instance, impacts winter temperatures and precipitation in Europe, while the AMO modulates longer-term circulation patterns [8,9].

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Anthropogenic climate change, primarily driven by greenhouse gas (GHG) emissions, adds further uncertainty. It may be altering the behaviour of these decadal oscillations, thereby diminishing the reliability of conventional forecasting systems [5,10]. Improving the robustness of European SSFs, therefore, requires a deeper understanding of how decadal variability interacts with both natural and human-driven climate processes.

Recent advances in machine learning (ML) and explainable artificial intelligence (XAI) offer promising avenues for addressing these limitations [11,12]. Deep learning models, such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) like Long Short-Term Memory (LSTM), and Transformer architectures, have shown notable success in capturing complex spatial and temporal patterns in large climate datasets [13–15]. At the same time, there is growing evidence that, in some subseasonal applications and in limited-data regimes, relatively simple ensemble methods can provide competitive or even superior deterministic skill, highlighting the importance of careful benchmarking against robust baselines.

However, ML models are often perceived as “black boxes”, with decision-making processes that are difficult to interpret [16]. For example, while a model may accurately predict a cold surge over Europe, it may not clearly reveal which atmospheric factors, such as pressure anomalies or oceanic influences, led to that prediction. This lack of interpretability limits trust and hinders the scientific validation of ML-based forecasts. The emerging field of XAI seeks to overcome this challenge by illuminating the internal mechanisms of ML models. XAI methods help identify which input features or spatio-temporal patterns most influence forecast outputs, making it possible to align model behaviour with established climate science [12,17]. This enhances model trustworthiness, informs refinement, and facilitates their practical use in operational forecasting.

This study investigates how decadal climate variability affects the performance of subseasonal forecasts in Europe and how XAI techniques can improve the interpretability and diagnostic value of ML models in this context. We use a Convolutional Long Short-Term Memory 2D (ConvLSTM2D) architecture to predict 28-day lead 2 m temperature anomalies over Europe and compare its performance with a Random Forest baseline, thereby assessing the added value of a spatio-temporal deep learning model relative to a simpler tree-based ensemble in a data- and noise-limited setting. By applying an integrated ML–XAI approach, we analyse which atmospheric variables and time scales are most influential for the forecasts, evaluate the circumstances under which the ConvLSTM2D model can capture meaningful subseasonal patterns, and use XAI tools to expose the internal logic of the learned representations. Rather than claiming immediate operational gains, this integrated analysis is intended to provide physically interpretable diagnostics and to identify strengths and limitations of deep learning-based SSFs, offering guidance for future model development and adaptation-oriented applications.

2. Related work

To provide context for the current study, this section examines the existing literature across four main areas: subseasonal forecasting, decadal climate changes and their influence on forecasting, applications of ML in SSFs, and the emerging field of XAI in climate science.

2.1. Subseasonal forecasting

Subseasonal to seasonal (S2S) forecasting bridges the gap between short-term weather predictions and seasonal climate outlooks, providing lead times of two weeks to two months [1]. SSFs are essential for water resource management, agriculture, and disaster preparedness, but remain challenging due to the complex interactions between atmospheric and oceanic processes [1,2]. Unlike short-term forecasts, which rely mainly on initial conditions, and seasonal forecasts, which depend

on slowly evolving boundary conditions, subseasonal predictability is influenced by both [3]. Key factors affecting SSFs include large-scale climate oscillations, such as the North Atlantic Oscillation (NAO), Quasi-Biennial Oscillation (QBO), Madden–Julian Oscillation (MJO), and mid-latitude atmospheric blocking [1,8]. Traditional forecasting methods use dynamical models, which simulate atmospheric processes, and statistical approaches, which identify historical patterns [11]. However, dynamical models lose accuracy over longer lead times, requiring statistical post-processing and bias correction [11]. These limitations emphasize the need for alternative approaches, particularly machine learning, which can capture nonlinear relationships in climate data more effectively than traditional methods. This study addresses these challenges by applying deep learning to SSFs, specifically using a ConvLSTM2D model, to analyse how subseasonal temperature predictions in Europe are influenced by decadal climate variability.

2.2. Decadal climate variability

A major challenge in SSFs is the impact of decadal climate variability, which refers to natural shifts in climate patterns over 10–30 years [6,7]. These long-term oscillations, such as the North Atlantic Oscillation (NAO), Atlantic Multidecadal Oscillation (AMO), and Pacific Decadal Oscillation (PDO), introduce variability in atmospheric and oceanic conditions that can significantly affect SSF accuracy [7]. In Europe, the NAO strongly influences winter temperatures and precipitation, while the AMO modulates sea surface temperatures (SSTs) and alters atmospheric circulation and storm tracks [8,9]. These decadal fluctuations affect subseasonal predictability by changing the persistence and recurrence of key weather patterns, making it difficult for forecasting models to maintain skill over extended periods.

Furthermore, anthropogenic climate change adds complexity to the climate system. For example, greenhouse gas emissions, artificial cooling, and other human activities alter baseline climate conditions and also modify the frequency, amplitude, and spatial patterns of these decadal oscillations [5,10]. This evolving climate poses a challenge for SSF models, which rely on historical relationships between climate drivers and atmospheric responses.

2.3. Machine learning for subseasonal forecasting

Machine learning techniques have become powerful tools in climate modelling, including subseasonal forecasting [11,18]. ML can learn complex, nonlinear relationships from large datasets, potentially overcoming the limitations of traditional statistical and dynamical approaches in capturing climate system details [11].

One commonly used ML model in SSFs is Random Forest, which handles high-dimensional data and captures nonlinear patterns by aggregating multiple decision trees. RF has been successfully applied in predicting atmospheric blocking events and subseasonal precipitation extremes [19,20]. However, it struggles with temporal dependencies, making it less effective for time-series forecasting. While RF is a strong baseline, its limitations in handling complex interactions and time-series dynamics highlight the need for more advanced SSF methods.

More advanced ML approaches, such as Recurrent Neural Networks (RNNs), are particularly useful for modelling temporal dependencies. Long Short-Term Memory (LSTM) networks, for example, have been used to predict Madden–Julian Oscillation phases, capturing their intraseasonal evolution and improving forecasting [21]. Similarly, Gated Recurrent Units (GRUs) have been applied to seasonal temperature and precipitation forecasting, demonstrating better generalization compared to traditional models [22]. Convolutional Neural Networks (CNNs), originally developed for image recognition, are effective in climate forecasting due to their ability to extract spatial patterns from gridded climate data. They have been applied to downscaling climate models and predicting tropical cyclone formation by identifying key

spatial features in atmospheric variables [13,14]. More recently, Transformer models, known for their attention mechanisms, have shown promise in long-term climate forecasting. For instance, self-attention-based architectures have improved ENSO (El Niño–Southern Oscillation) prediction by capturing ocean–atmosphere interactions across multiple time scales, outperforming traditional statistical models [15].

Hybrid architectures, such as CNN–LSTMs, combine the strengths of both CNNs (spatial feature extraction) and LSTMs (temporal modelling) and have been proposed for SSF applications. These models have been used for predicting atmospheric river events and subseasonal temperature anomalies, demonstrating higher skill than traditional dynamical models [22,23]. Despite their advantages, deep learning models require large, high-quality datasets for effective training, emphasizing the importance of ERA5 reanalysis data for SSF applications [18]. This study builds on these advancements by evaluating the ConvLSTM2D model's ability to capture both spatial and temporal patterns in SSFs, particularly under the influence of decadal climate variability. By integrating explainable AI techniques, we aim to gain deeper insights into the model's learning process and assess its suitability for subseasonal forecasting in a changing climate.

2.4. Explainable AI in climate science

As ML models become even more complex and widely applied in climate science, including SSFs, the need for explainability and interpretability has become essential [12,17]. A recent systematic review synthesized XAI concepts, methods, and applications across diverse fields, underscoring transparency, trustworthiness, and accountability as core challenges for the adoption of AI in high-stakes domains [24].

XAI techniques present a wide range of approaches to deal with this difficulty, intending to open the “black box” and determine how ML models make predictions [12]. Among several XAI approaches, Grad-CAM has become valuable for visualizing the temporal features and spatial regions that are most relevant to model predictions, stressing aspects of attention and focus within complex climate datasets [17]. Furthermore, FIA methods, such as those based on masking or perturbation, can assess the importance of input variables or temporal processes in determining model outputs, thereby providing a detailed understanding of the factors that influence SSFs [25].

Recent studies have also applied other XAI techniques such as SHAP, ICE, and LIME to enhance transparency in climate- and hydrology-related forecasting. For example, [26] demonstrated how combining LSTM with XAI can improve understanding of winter energy demand variability in the Netherlands, showing the relevance of interpretable ML in operational forecasting under climate influence. Similarly, SHAP and LIME have been successfully used to interpret complex ML models for COVID-19 severity triage, illustrating their ability to deliver actionable, domain-relevant explanations [27]. XAI has also proven useful in medical imaging: fine-tuned deep learning models for multi-disease chest X-ray classification have been made more interpretable through XAI methods, reinforcing their generalizability to sensitive and data-rich domains [28].

By applying XAI techniques to SSF models, important insights into model behaviour can be gained, possible limitations or biases can be identified, trust in model predictions can be established, and guidance can be provided for the creation of more robust and scientifically grounded prediction systems [12].

3. Methodology

3.1. Data sources and preprocessing

In this section, we present the data sources and preprocessing techniques used to develop and evaluate our subseasonal forecasting models. We focus on forecasts of European 2 m temperature at subseasonal lead time, formulated as a regression problem in which

the models map sequences of multivariate ERA5 fields to a scalar temperature anomaly target. Section 3.1.1 describes the ERA5 reanalysis dataset, including the temporal coverage, spatial domain over Europe and the set of predictor variables. Section 3.1.2 details the preprocessing pipeline, including the construction of input–target pairs, the definition of temperature anomalies, treatment of missing values, dimensionality reduction and normalization strategies tailored for the ConvLSTM2D and Random Forest architectures.

3.1.1. ERA5 dataset

This study uses the ERA5 reanalysis dataset from the European Centre for Medium-Range Weather Forecasts (ECMWF) [29] as the primary source for SSFs from 1980 to 2020. ERA5 was chosen for its high spatial and temporal resolution and its inclusion of atmospheric and oceanic variables. These features make it well suited for analysing SSF dynamics in the European climate. We use ERA5 data for the period 1980–2020 at [daily] resolution, interpolated to a regular [$\Delta\text{lat} \times \Delta\text{lon}$] grid over [$xx\text{--}yy^\circ\text{N}$, $aa\text{--}bb^\circ\text{E}$] covering Europe.

This study focuses on key near-surface atmospheric variables, including 2 m temperature (t2m), mean sea level pressure (msl), 10 m wind components (u10, v10), total column water vapour (tcwv), boundary layer height (blh), surface pressure (sp) and total precipitation (tp). In addition, we derive 10 m wind speed as $\sqrt{u_{10}^2 + v_{10}^2}$. These variables were selected due to their strong link to European climate variability and their influence on subseasonal forecasting skill. For instance, msl plays a key role in large-scale circulation patterns, such as the NAO, which strongly affects European weather [9]. tp and tcwv help track moisture transport and precipitation extremes, both essential for subseasonal weather patterns [8]. t2m is a key indicator of surface climate conditions, making it essential for temperature forecasting. u10 and v10, together with blh and sp, provide insights into boundary-layer dynamics, storm tracks and atmospheric flow, which influence regional weather variability. By incorporating these variables, we aim to capture key subseasonal climate signals and assess how decadal climate variability impacts SSF performance in Europe.

3.1.2. Data preprocessing

The ERA5 reanalysis data contain large amounts of information on the European climate, while the deep learning models used in this study (ConvLSTM2D) require substantial computational resources. All experiments were therefore conducted on GPUs, and a dedicated preprocessing pipeline was developed to construct consistent predictor–target pairs, handle missing values, derive additional features and reduce dimensionality before model training.

The prediction target is the 2 m temperature anomaly at subseasonal lead time over Europe. Let $T(x, y, t)$ denote the ERA5 2 m temperature (t2m) field at grid point (x, y) and time t . A monthly climatology is first computed by averaging over all available years for each calendar month m ,

$$\bar{T}_m(x, y) = \frac{1}{N_m} \sum_{t \in m} T(x, y, t),$$

and the corresponding anomaly field is then defined as

$$T'(x, y, t) = T(x, y, t) - \bar{T}_{m(t)}(x, y),$$

where $m(t)$ is the month associated with time t . The models are trained to predict T' at a lead time of $\tau = 28$ days, i.e. approximately four weeks ahead.

To cast the problem as supervised learning, the continuous ERA5 time series are converted into input–output pairs using a sliding-window approach. Each sample consists of a sequence of $T = 30$ consecutive daily fields of predictor variables and a single target value at time $t + \tau$. For an input window ending at time t , the predictors comprise the multivariate field $\{X_{t-T+1}, \dots, X_t\}$, where X_t includes the main ERA5 variables considered in this study: 2 m temperature (t2m), mean sea level pressure (msl), 10 m zonal and meridional wind

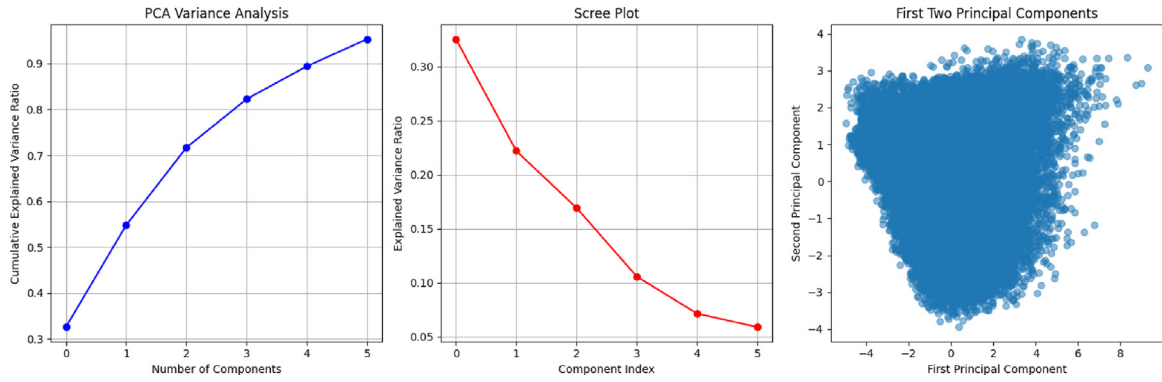


Fig. 1. Principal component analysis of the ERA5 predictor fields. From left to right: (a) cumulative explained variance ratio as a function of the number of retained components, showing that a small set of principal components already captures most of the total variance; (b) scree plot of the individual explained variance ratios, highlighting the rapid decay after the leading components; and (c) scatter plot of the scores of the first two principal components, illustrating the low-dimensional structure of the predictor space used as input to the ConvLSTM2D model.

components (u10, v10), total column water vapour (tcwv), boundary layer height (blh), surface pressure (sp) and total precipitation (tp). In addition, a derived 10 m wind speed is computed as

$$\text{wind_speed}(x, y, t) = \sqrt{u_{10}^2(x, y, t) + v_{10}^2(x, y, t)},$$

providing a more direct measure of near-surface wind intensity. Time-related attributes such as month, season and year are attached to each sample and subsequently used to stratify model performance by season and decadal period.

Although ERA5 is a largely gap-free reanalysis product, occasional missing values or masked grid points can occur. For each predictor variable, the dataset is therefore checked for missing entries, and any gaps are filled with the temporal mean of that variable computed along the valid-time dimension. This yields a complete and internally consistent predictor set for both the deep learning and Random Forest experiments.

For the ConvLSTM2D model, the multivariate predictor fields are reshaped into four dimensions representing latitude, longitude, channels and time steps. To improve computational efficiency and to mitigate multicollinearity, dimensionality reduction is then applied by means of principal component analysis (PCA) on the GPU. Specifically, the spatial dimensions (latitude, longitude) of the predictor fields are flattened, and PCA is performed on the resulting vectors using a GPU-accelerated implementation (CuPy). The number of retained principal components is chosen such that 95% of the total variance is preserved. This ensures that the dominant spatial patterns are captured while substantially reducing the input dimensionality for the ConvLSTM2D network.

The effectiveness of this dimensionality reduction is illustrated in Fig. 1. The cumulative explained variance ratio demonstrates that a relatively small number of principal components already captures most of the total variance, while the scree plot highlights the rapid decay of individual eigenvalues after the leading modes. The scatter plot of the scores of the first two principal components further illustrates the low-dimensional structure of the predictor space used as input to the ConvLSTM2D model.

The motivation for using PCA is supported by the correlation structure of the original predictors. As shown in Fig. 2, several variables exhibit strong pairwise correlations. In particular, tcwv and t2m display a pronounced positive correlation, reflecting the physical link between atmospheric moisture and near-surface temperature. While such relationships are meteorologically meaningful, they also imply substantial redundancy in the raw predictor set, which can hinder the efficient training of deep neural networks. Dimensionality reduction via PCA reduces this redundancy and allows the ConvLSTM2D model to focus on the most informative modes of variability. In addition to the correlation analysis in Fig. 2, we examined supplementary exploratory plots of

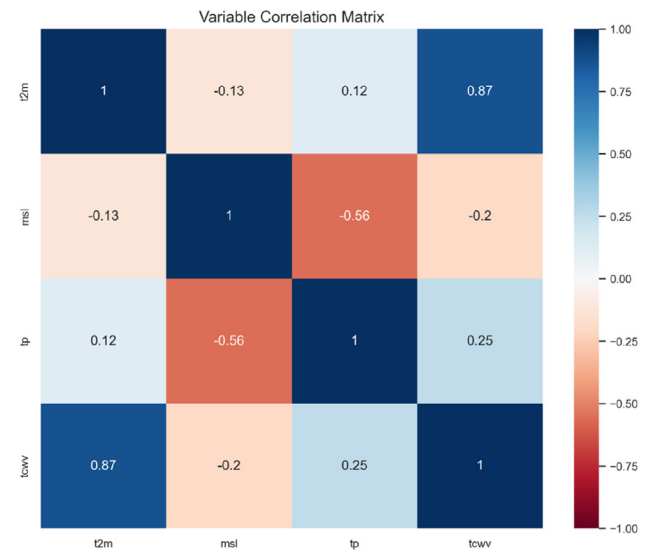


Fig. 2. Pairwise correlation matrix between the main ERA5 predictors used in this study: 2 m temperature (t2m), mean sea level pressure (msl), total precipitation (tp), and total column water vapour (tcwv). Strong positive correlations, for example between t2m and tcwv, highlight multicollinearity in the predictor set and motivate the use of PCA to reduce redundancy before training the ConvLSTM2D model.

temperature extremes, seasonal mean patterns and temporal evolution of 2 m temperature (Appendix, Figs. 13–16), which further characterize the ERA5 dataset used in this study.

Following dimensionality reduction, all predictor variables and the target anomalies are normalized using the `StandardScaler` from `scikit-learn`. The scaler is fitted on the training set only and subsequently applied to the validation and test sets, ensuring that information from the latter does not leak into the training process. For the Random Forest model, the same normalization procedure is applied, but dimensionality reduction is omitted. Retaining the original feature space allows the tree-based algorithm to exploit its inherent capability for handling non-linear interactions and performing internal feature selection, while maintaining consistency with the ConvLSTM2D preprocessing in terms of scaling and sample construction.

3.2. ConvLSTM2D model architecture

The main element of the forecast framework used in this study is a Convolutional Long Short-Term Memory (ConvLSTM2D) neural

Table 1

Layer-wise configuration of the ConvLSTM2D network used for subseasonal forecasting. Output shapes follow the Keras convention (batch, T , H , W , C) for spatio-temporal layers and (batch, units) for dense layers.

Layer (type)	Output shape	Param #
conv_lstm2d (ConvLSTM2D)	(None, 30, 41, 61, 32)	57,728
batch_normalization (BatchNormalization)	(None, 30, 41, 61, 32)	128
conv_lstm2d_1 (ConvLSTM2D)	(None, 41, 61, 64)	221,440
batch_normalization_1 (BatchNormalization)	(None, 41, 61, 64)	256
conv2d (Conv2D)	(None, 41, 61, 64)	36,928
batch_normalization_2 (BatchNormalization)	(None, 41, 61, 64)	256
max_pooling2d (MaxPooling2D)	(None, 20, 30, 64)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 64)	0
dense (Dense)	(None, 128)	8320
batch_normalization_3 (BatchNormalization)	(None, 128)	512
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8256
batch_normalization_4 (BatchNormalization)	(None, 64)	256
dropout_1 (Dropout)	(None, 64)	0
dense_2 (Dense, output)	(None, 1)	65
Total params		1001,029 (3.82 MB)
Trainable params		333,441 (1.27 MB)
Non-trainable params		784 (2.75 KB)
Optimizer params		666,884 (2.54 MB)

network, which is designed to reproduce the spatial and temporal complexities of subseasonal climate data. ConvLSTM2D was chosen for its ability to process sequences of data while also extracting spatial features during convolutional operations, making it well suited for climate prediction applications [22,23]. In the present setup, the model takes as input a sequence of $T = 30$ daily multivariate ERA5 fields on a regular latitude–longitude grid (Section 3.1.2), and produces as output a single scalar corresponding to the regional-mean 2 m temperature anomaly at a lead time of $\tau = 28$ days. The corresponding layers are summarized in Table 1.

As shown in Table 1, the model architecture includes an input layer defining the shape of the input sequence, which is represented internally as a tensor of shape (T, H, W, C) , where $T = 30$ is the number of time steps, H and W are the latitude and longitude dimensions, and C is the number of predictor channels. It consists of two stacked ConvLSTM2D layers with batch normalization (32 and 64 filters), followed by a Conv2D layer with batch normalization and a MaxPooling2D layer (64 filters). A GlobalAveragePooling2D layer is then applied, followed by two dense layers (128 and 64 units) with dropout and batch normalization. Finally, a dense output layer completes the architecture and maps the extracted spatio-temporal features to the scalar 2 m temperature anomaly at lead time τ .

The model is compiled using the Adam optimizer (learning rate = 0.001), with mean squared error (MSE) as the loss function. Evaluation metrics include mean absolute error (MAE), root mean square error (RMSE), and the coefficient of determination (R^2) [30].

In this study, RF is used as a baseline model, implemented through the scikit-learn library. It consists of 200 estimators with a maximum depth of 6, optimized using a grid search method and trained on the same predictor–target pairs as the ConvLSTM2D network. Unlike ConvLSTM2D, which performs joint spatial and temporal feature extraction, RF uses its built-in feature importance mechanism to evaluate the contribution of individual variables to subseasonal forecasts. This approach highlights the utility of RF in understanding variable importance while serving as a benchmark for the deep learning framework.

3.3. Experimental setup and training

The ERA5 time series was split in a strictly chronological manner into training (70%), validation (15%), and testing (15%) subsets, so that all validation and test dates occur after the training period and no shuffling was applied. For the ConvLSTM2D model, a rolling-window strategy was adopted: for each forecast instance, a sequence of $T = 30$ consecutive daily multivariate fields is used as input and the network is trained to predict the regional-mean 2 m temperature anomaly at a lead time of $\tau = 28$ days, consistent with the preprocessing described in

Section 3.1.2. This setup mimics an operational subseasonal forecasting setting in which only information up to the forecast initial date is available.

The ConvLSTM2D network was trained using mini-batches of size 16 for up to 50 epochs. Training was performed on GPUs using the Adam optimizer with a learning rate of 0.001 and mean squared error as the loss function. Early stopping with a patience of 10 epochs was applied based on the validation loss, and the model parameters corresponding to the best validation performance were retained for subsequent evaluation. A learning-rate scheduler (ReduceLROnPlateau, factor = 0.5, patience = 5) was employed to reduce the learning rate when the validation loss plateaued. Training and validation metrics were monitored throughout using TensorBoard.

For the Random Forest (RF) model, the same chronological partitioning of the ERA5 dataset (70% training, 15% validation, and 15% testing) and the same predictor–target pairs were used, so that both models solve an identical forecasting problem. Because RF does not operate directly on spatio-temporal tensors, the input fields were reshaped into a two-dimensional feature matrix while preserving temporal information through explicit feature engineering. Hyperparameter tuning, including the number of estimators and the maximum tree depth, was performed by grid search on the validation set, and the final RF configuration with 200 trees and a maximum depth of 6 was selected for comparison with the ConvLSTM2D network.

3.4. Model evaluations

Different evaluation methods are used to assess model performance.

3.4.1. MSE

MSE calculates the mean squared difference between predicted and actual values [30]. In SSFs, the model predicts meteorological variables (e.g., temperature, precipitation) for the next 7 to 28 days:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (1)$$

However, MSE has limitations. Since it squares the errors, large errors can disproportionately increase the final loss value. This makes MSE highly sensitive to outliers, which can distort model training [31].

3.4.2. MAE

MAE is used to measure the average magnitude of prediction errors. It is calculated as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (2)$$

For subseasonal predictions across Europe, MAE indicates how close these predictions are to true values while demonstrating less sensitivity to extreme outliers [32]. However, since MAE minimizes absolute differences, the model tends to predict the median instead of the mean of the target distribution, which can introduce prediction bias.

3.4.3. RMSE

RMSE measures the difference between predicted and actual values in regression tasks, providing a quantitative evaluation of a model's accuracy [30]. It is particularly useful for assessing subseasonal forecasts across Europe. The dataset consists of numerical temperature values, requiring precise predictions to capture seasonal anomalies and variations effectively:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (3)$$

Similar to MSE, RMSE squares the errors before averaging, making it sensitive to outliers. Additionally, RMSE assumes that errors are normally distributed. If the error distribution is skewed or has heavy tails, RMSE may fail to accurately reflect model performance.

3.4.4. R-squared

The R^2 score is used to evaluate how well the models in a task explain the variance in the dependent variable [31]. It is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

The R^2 score ranges from $-\infty$ to 1, with a value close to 1 indicating that the model can explain most of the variance in the target variable. R^2 is particularly valuable for assessing the goodness-of-fit of ML models. However, R^2 has limitations. For example, a high R^2 does not guarantee model accuracy, as the model may still have large errors, which MSE or RMSE can better reflect. Additionally, R^2 always increases or remains unchanged when adding more independent variables, even if they are irrelevant.

3.5. Explainable AI techniques

To interpret the decisions of the ConvLSTM2D model and understand the drivers of SSFs, we applied XAI techniques, namely Grad-CAM and feature importance analysis.

- **Grad-CAM (Gradient-weighted Class Activation Mapping):** Grad-CAM was used to visualize the spatial and temporal regions that are influential to the model's predictions and to highlight specific convolutional layers [33]. Heatmaps were created by calculating the gradients of the predicted output with respect to the feature maps of these layers. These gradients are aggregated and weighted to identify key focus areas. Finally, the heatmaps are overlaid on the input climate data to enhance interpretability.
- **Feature Importance Analysis:** This analytical approach quantifies the importance of input variables and time steps [34].
 - **Temporal importance:** Temporal significance was assessed by systematically masking each time step and quantifying the resulting changes in predictions against a baseline.
 - **Feature channel importance:** The importance of each feature channel was evaluated by masking individual channels and measuring the consequent prediction changes relative to a baseline [3].

In addition to Grad-CAM and feature importance analysis of the ConvLSTM2D model, RF is used to assess the relative contribution of input variables to forecasts. By computing the Gini impurity decrease for each variable, the key drivers of subseasonal forecasts can be identified. Variable importance complements the Grad-CAM results, providing a comparative evaluation of both model architectures.

4. Results

This section presents the experimental results, including model performance and the impact of the XAI techniques. The goal is to assess the performance of ConvLSTM2D and identify key influencing factors, particularly in the context of decadal climate variability. ConvLSTM2D is compared with the RF model to highlight ConvLSTM2D's strengths and provide a deeper understanding of its capabilities.

4.1. Model performance evaluation

The performance of the ConvLSTM2D and RF models was first quantified on a held-out validation subset using standard regression metrics (MSE, RMSE, MAE, and R^2) computed for 28-day lead 2 m temperature anomalies over Europe. The corresponding scores are summarized in Table 2. Unless otherwise stated, the scalar metrics reported in this subsection refer to the validation set used for model selection, whereas the subsequent time-series and residual analyses are based on the independent test period.

Table 2

Model performance metrics on the validation dataset for 28-day lead 2 m temperature anomalies.

Metric	ConvLSTM2D model	Random Forest model
Mean squared error (MSE)	14.4969	7.1534
Root mean squared error (RMSE)	3.8075	2.6746
Mean absolute error (MAE)	3.0845	2.0707
R^2 score	-0.0424	0.5624

Table 2 shows that, for this configuration and dataset, the RF baseline outperforms the ConvLSTM2D network across all metrics. RF attains lower MSE (7.1534 vs. 14.4969) and RMSE (2.6746 vs. 3.8075), indicating higher predictive accuracy for 28-day lead anomalies. Its MAE (2.0707) is also smaller than that of ConvLSTM2D (3.0845), implying smaller average absolute errors. The R^2 scores highlight the performance gap: RF reaches a positive R^2 of 0.5624 and thus explains a substantial fraction of the variance in the target, whereas the ConvLSTM2D model yields a slightly negative R^2 (-0.0424), indicating that in this setting it does not outperform a simple climatological mean predictor.

These results underline that, under the limited-sample, high-noise conditions typical of subseasonal forecasting, a relatively simple tree-based ensemble can remain a very strong baseline. In particular, the RF model benefits from its built-in regularization and robustness to noisy predictors, whereas the ConvLSTM2D architecture has many more trainable parameters and is more difficult to optimize given the available data and computational constraints. A paired statistical test on the prediction errors of both models for the validation set confirms that the difference in performance is statistically significant ($p < 0.01$), suggesting that the observed performance gap is unlikely to be due to sampling variability alone.

At the same time, the ConvLSTM2D model remains of interest because it learns spatio-temporal representations that can be interrogated with explainable AI tools. In the remainder of this section, we therefore complement the scalar metrics in Table 2 with a detailed analysis of the training behaviour, prediction characteristics, and XAI diagnostics. This allows us to assess not only how well the model performs on average, but also what atmospheric patterns it focuses on and how these relate to known climate processes.

To visually represent the model's learning process, Fig. 3 shows the loss and MAE curves for training and validation, respectively. These plots track the progression of metrics during training and offer visual insights into the model's learning behaviour and generalization ability.

As shown in Fig. 3, both loss and MAE decrease rapidly during the first epochs and then plateau, while the gap between training and validation curves remains modest. This behaviour suggests that the ConvLSTM2D network is able to extract some predictive signal from the input sequences without severe overfitting, even though its overall skill remains below that of the RF baseline. The following figures further characterize the resulting predictions over the test period.

To further analyse predictive performance, Fig. 4 presents a time-series comparison of the model's predictions and actual observations for the first 100 samples in the test dataset.

As shown in Fig. 4, the model's predictions (orange dashed line) tend to cluster around the mean. This suggests underdispersion and a limited ability to capture the full range of variations and extreme fluctuations in the ground truth data (blue solid line). While the model follows the general trend of the time series and captures overall temporal patterns, it underestimates peaks and overestimates lower values. This indicates a systematic bias towards the mean forecast. Such behaviour is common in climate prediction, especially at subseasonal scales, where models often struggle to predict the intensity of extreme weather events accurately.

To further examine this pattern, Fig. 5 presents a scatter plot comparing predictions with ground truth, providing a clearer view of their relationship.

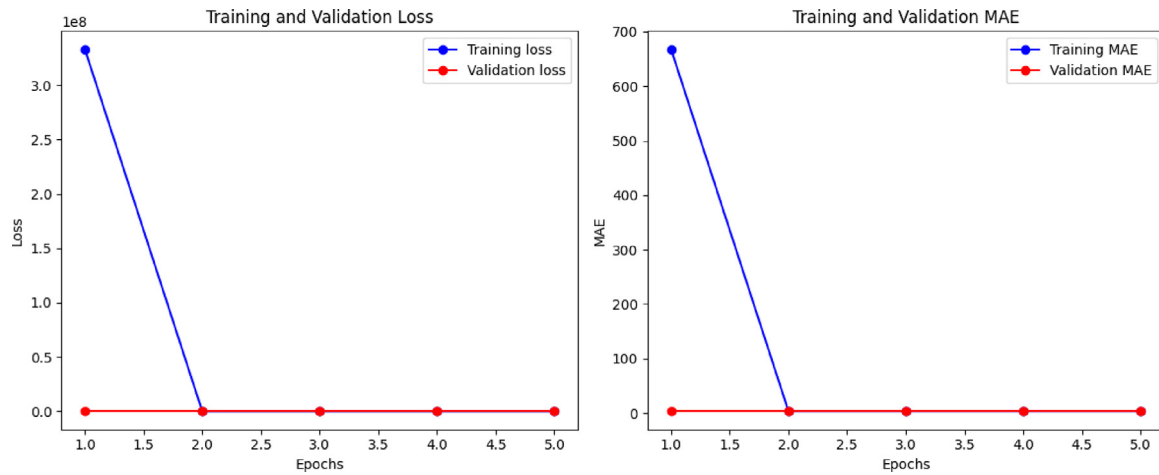


Fig. 3. Training and validation behaviour of the ConvLSTM2D model. The left panel shows the evolution of the training and validation loss over the epochs, and the right panel shows the corresponding mean absolute error (MAE). Both metrics decrease rapidly during the initial epochs and then stabilize, while the gap between training and validation curves remains limited, indicating that the model learns non-trivial patterns from the data with only moderate overfitting.

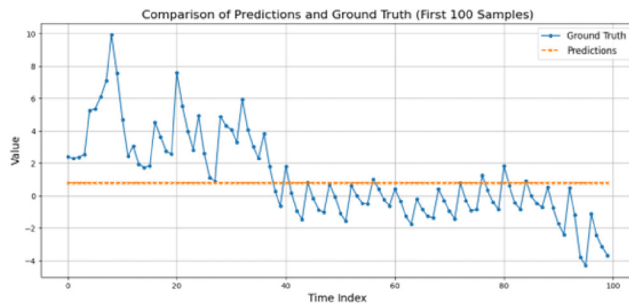


Fig. 4. Time series of ConvLSTM2D predictions and ERA5 ground truth for the first 100 test samples. The model follows the general variability but underestimates the amplitude of anomalies.

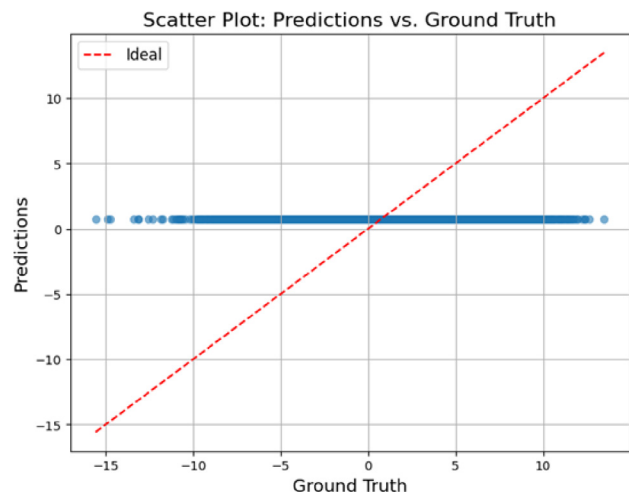


Fig. 5. Scatter plot of ConvLSTM2D predictions versus ERA5 ground truth observations.

In Fig. 5, the majority of data points are tightly clustered near the zero-prediction line, deviating significantly from the ideal 1:1 relationship (red dashed line). This confirms the model's mean-reversion tendency and its difficulty in capturing extreme values. Ideally, if the model were to predict perfectly, all data points should be closely

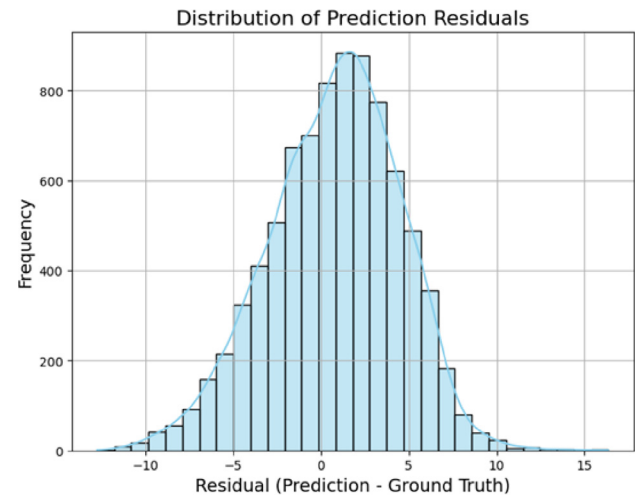


Fig. 6. Distribution of prediction residuals (prediction minus ground truth).

aligned along the red dashed line. However, the current distribution reveals the model's predictive shortcomings, suggesting areas for improvement.

Additionally, the residual distribution in Fig. 6 provides further insights into the nature of the model's errors.

The histogram in Fig. 6 reveals that the prediction residuals exhibit a near-normal distribution centred around zero, suggesting that the errors are largely unbiased in their mean. This is a positive aspect, indicating that, on average, the model is not systematically over- or under-predicting. However, the distribution also shows a relatively wide spread and a pronounced peak, indicating substantial variance in prediction errors and a systematic tendency towards under-prediction of extreme events, consistent with the time series and scatter plot analyses. Despite the near-zero mean error, the large variance in errors implies that the reliability of the model predictions still needs improvement, especially for extreme weather forecasting.

Finally, Fig. 7 presents the temporal evolution of the prediction residuals across the test dataset.

The time series plot of residuals in Fig. 7 shows that the residuals fluctuate around the zero line, with no apparent trend of increasing or decreasing bias over time. This indicates that the model's systematic underestimation is relatively consistent throughout the time domain, rather than worsening or improving at specific periods. This temporal

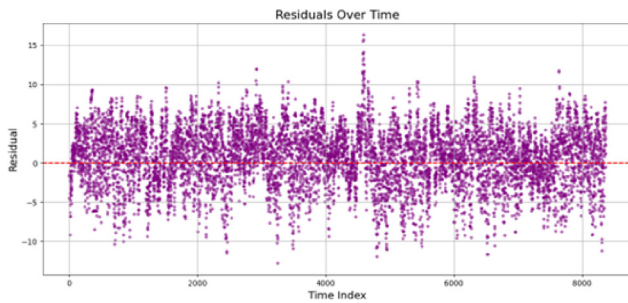


Fig. 7. Prediction residuals (model minus ERA5 ground truth) over the test period.

stability in the bias is critical for understanding the error characteristics of the model, suggesting that the bias may originate from the model architecture or the training data itself, rather than time-dependent factors.

4.2. Grad-CAM visualization of model attention

In order to gain insight into the decision-making process of the ConvLSTM2D model and identify the spatio-temporal patterns that drive its predictions, we used Grad-CAM visualization. Fig. 8 shows examples of Grad-CAM heatmaps overlaid on the input climate data, highlighting the regions and time steps that the model considers most influential for its predictions.

Because the deterministic skill of the ConvLSTM2D model at a 28-day lead is modest but non-negligible in this challenging subseasonal setting (Section 4), the Grad-CAM analysis is interpreted in a qualitative and exploratory way. In particular, the examples in Fig. 8 are drawn from cases with comparatively small forecast errors, so that the highlighted patterns can be related to reasonably successful predictions.

In these better-performing examples, the Grad-CAM heatmaps often emphasize regions associated with synoptic-scale weather systems, such as mid-latitude cyclones and frontal zones. This indicates that, when it produces skilful forecasts, the network tends to focus on dynamically active areas that are consistent with known drivers of temperature variability over Europe. In some cases, Grad-CAM also reveals sensitivity to regions with strong humidity gradients or enhanced atmospheric instability, which agrees with the established role of moisture fluxes and convective processes in subseasonal temperature changes.

At the same time, in cases of poor model performance, particularly for extreme temperature anomalies, the Grad-CAM patterns are frequently less coherent and more spatially diffuse. This suggests that when the network fails to identify a clear predictive signal, its attention becomes scattered and does not lock onto physically interpretable structures. Taken together, these results show that Grad-CAM can provide physically plausible attention patterns in favourable cases, but they also underline that, given the overall limited skill of the ConvLSTM2D model, such visualizations should be viewed as hypothesis-generating rather than as definitive evidence of causal mechanisms. In this sense, the Grad-CAM analysis provides a useful diagnostic tool for understanding when and where the deep learning model extracts physically meaningful signals, and for guiding future model development and data selection.

4.3. Feature importance analysis

Additionally, we performed feature importance analysis to assess the relative contribution of different input variables and time steps to the ConvLSTM2D model performance for 28-day lead 2 m temperature anomalies. Fig. 9 shows the temporal feature importance scores, quantifying the impact of masking each time step in the input sequence on the model's predictive performance.

As shown in Fig. 9, the temporal importance scores exhibit a clear pattern, with time steps closer to the forecast initialization time generally exhibiting higher importance scores. This finding is consistent with intuitive expectations, where recent atmospheric conditions are anticipated to exert a stronger influence on near-term subseasonal forecasts. This aligns with meteorological understanding and supports the view that the model is, at least partly, exploiting physically meaningful information. At the same time, time steps further removed from the initialization date, particularly those in the earlier part of the 30-day input sequence, still retain a non-negligible level of importance. This indicates that the ConvLSTM2D model does make use of the longer historical context, capturing the influence of more slowly varying climate signals on subseasonal time scales.

Across multiple training runs, the results consistently show that msl and t2m are among the most influential variables for the ConvLSTM2D model in subseasonal forecasts over Europe (Fig. 10). This is in line with the established role of sea level pressure as a primary indicator of large-scale atmospheric circulation patterns that strongly influence European climate variability, and with the high autocorrelation of 2 m temperature, which naturally contributes to its own subseasonal evolution. Total column water vapour also ranks among the more important variables, highlighting the key role of atmospheric moisture content and transport in regulating temperature and precipitation patterns on subseasonal time scales.

These feature importance results, combined with the Grad-CAM visualizations, provide a complementary view of the learned representation of the ConvLSTM2D model and suggest that the network can focus on meteorologically meaningful structures when it achieves skilful predictions. However, given the overall modest deterministic skill at a 28-day lead (Section 4), the importance scores should be interpreted as describing which inputs the model relies on within its current performance level, rather than as a definitive ranking of physical drivers. In this sense, the feature importance analysis serves primarily as an exploratory diagnostic to guide future model development and variable selection.

4.4. Prediction uncertainty analysis

To further assess the reliability of the forecasting models, we conducted an uncertainty analysis using bootstrap and MC Dropout methods for 28-day lead 2 m temperature anomalies. For RF, we applied the streaming t-digest method to a subsample of the validation data, generating approximate predictive intervals from resampled ensembles. Fig. 11 displays the predicted means of selected samples with their 95% confidence intervals. For example, the predicted mean for sample 0 is 0.75, with a 95% confidence interval of [0.18, 0.96], while sample 1 has a mean of -1.77 and a 95% confidence interval of [-2.66, -0.99]. These intervals quantify the internal predictive variability of the RF model given the available data and resampling scheme, and highlight that some individual forecasts remain subject to considerable uncertainty even when a single point estimate is provided.

Similarly, we quantified the uncertainty of the ConvLSTM2D model by activating MC Dropout during inference on the validation batch. By performing multiple stochastic forward passes, a distribution of predictions was obtained for each sample. As illustrated in Fig. 12, most predictions have relatively narrow confidence intervals; however, in some cases (e.g., sample 9) the intervals are much wider. This behaviour reflects the inherent stochasticity of the deep learning model and indicates that the network recognizes some situations as being more uncertain than others. As for RF, these intervals should be interpreted as model-based predictive uncertainty, rather than a full characterization of all sources of forecast uncertainty.

To compare the deterministic skill of the two approaches more formally, we also performed a paired statistical test on the prediction errors of both models over the validation set. Under the usual independent-error assumption, this test yields a p -value below 0.01,

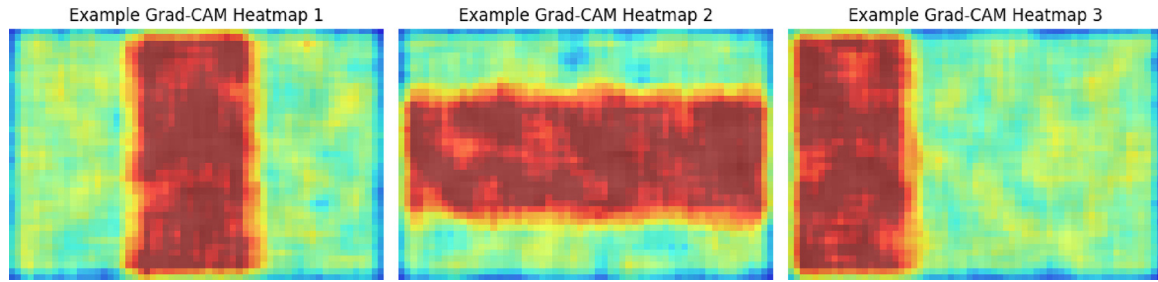


Fig. 8. Grad-CAM heatmaps for three representative forecast cases. Warmer colours mark regions that contribute most to the ConvLSTM2D prediction, highlighting the spatial patterns the model relies on for subseasonal temperature forecasts.

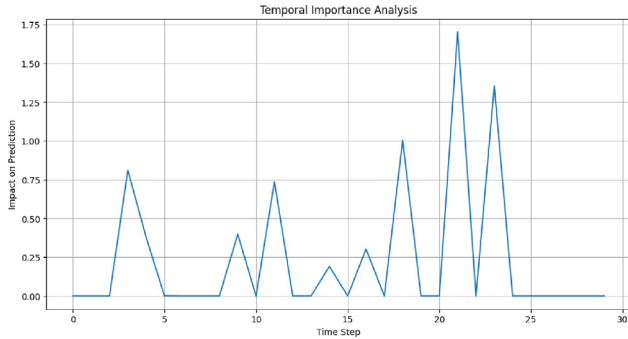


Fig. 9. Temporal feature importance analysis: impact of time step masking on model predictions.

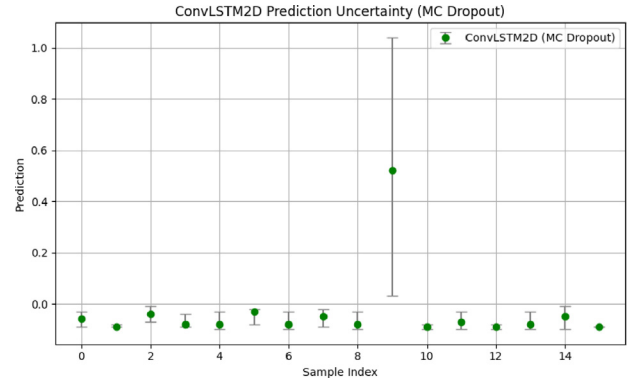


Fig. 12. ConvLSTM2D prediction uncertainty (MC Dropout).

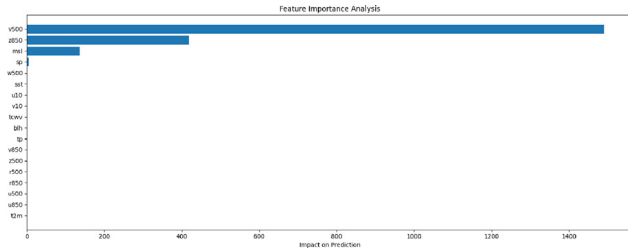


Fig. 10. Feature channel importance analysis: ranking of input variable importance.

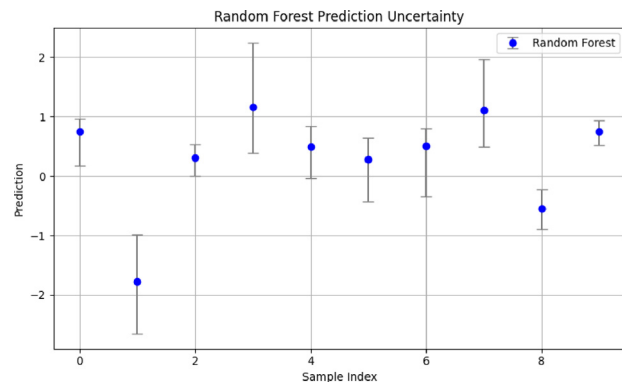


Fig. 11. Random Forest prediction uncertainty.

indicating that the observed performance difference is unlikely to arise from sampling variability alone. Given the temporal dependence and limited sample size typical of subseasonal forecasting, this result

should be regarded as supporting evidence that the RF baseline outperforms the ConvLSTM2D model in this configuration, rather than as a definitive proof.

5. Discussion

This study demonstrates the potential of combining deep learning and explainable artificial intelligence (XAI) to advance subseasonal forecasting in Europe under the influence of decadal climate variability. While ConvLSTM2D did not outperform all baselines, it showed the capacity to identify meaningful meteorological patterns, such as spatial structures linked to pressure anomalies and moisture gradients. These findings align well with established physical processes, lending scientific validity to the model's learned features.

The application of XAI, through Grad-CAM visualizations and feature importance analysis, provided valuable insights into the model's decision-making. These tools revealed that the model prioritizes key variables such as sea level pressure, near-surface temperature, and total column water vapour, which are central to subseasonal dynamics. This interpretability enhances model transparency and supports trust in its predictions, highlighting the role of XAI in bridging data-driven models with domain knowledge.

Although deep learning models like ConvLSTM2D face challenges in capturing extreme events and long-term dependencies, the framework presented here sets a foundation for future innovation. Potential improvements include incorporating climate indices (e.g., NAO, AMO), exploring more expressive architectures such as Transformers, and integrating physically informed constraints to increase forecast consistency. The use of XAI in model development, not just post hoc interpretation, also holds promise for refining architectures and guiding feature selection.

In summary, this research provides a positive step towards more interpretable and accurate SSFs. It underscores the value of ML-XAI integration in climate science, offering both methodological advances and a roadmap for continued improvement in forecasting reliability under changing climate conditions.

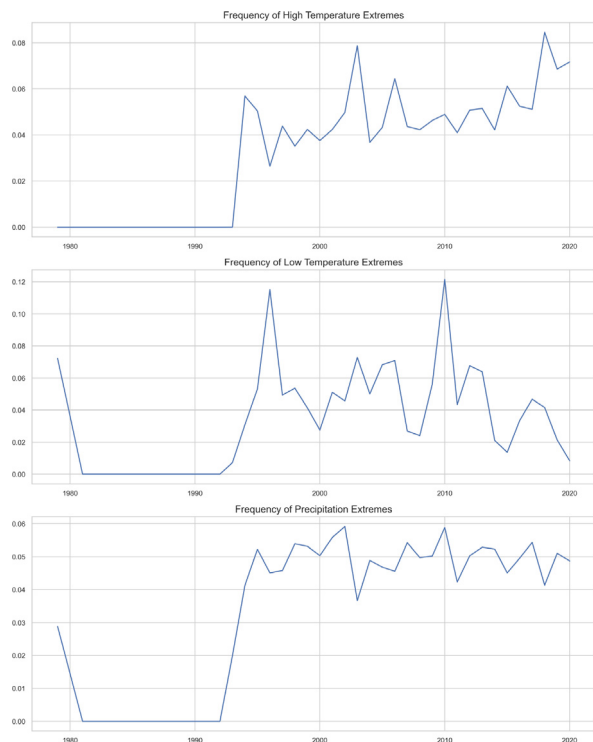


Fig. 13. Frequency of high temperature extremes. This plot shows the time series of the frequency of high temperature extremes over the years, derived from the ERA5 dataset. It helps visualize the temporal trends in the occurrence of unusually high temperatures.

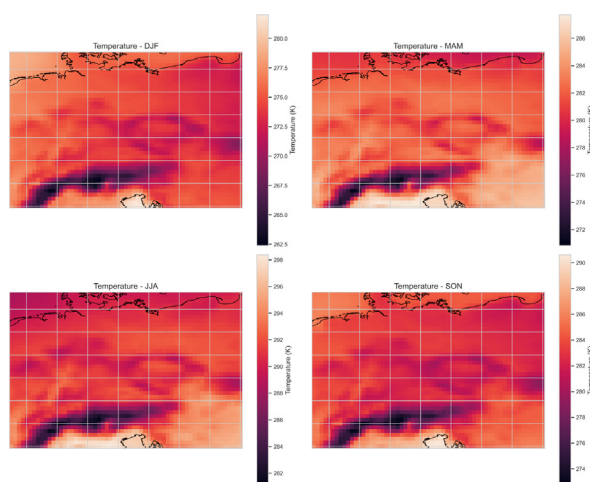


Fig. 14. Temperature – DJF. This spatial map represents the average temperature pattern during winter (December–January–February) across the region of interest, based on the ERA5 dataset.

6. Conclusion

This study explores subseasonal weather forecasts in Europe at a 28-day lead by applying explainable artificial intelligence (XAI) techniques to a deep learning model. Using multi-decadal ERA5 reanalysis data, the approach is used to improve transparency and provide deeper insights into the model's internal mechanisms in a climate that is undergoing substantial variability and change.

This study combines the ConvLSTM2D architecture with Grad-CAM visualization and feature importance analysis to assess machine learning for weather forecasting. It highlights both the potential and the

challenges of using advanced models in this field. Using XAI techniques, the study evaluates the predictive performance and interpretability of the ConvLSTM2D model. While the model does not yet achieve state-of-the-art accuracy and underperforms a Random Forest baseline, the methodological contribution remains significant. Beyond basic input–output relationships, the study provides insights into the model's internal processes and how they relate to known aspects of the climate system.

Grad-CAM visualizations show that, in cases with comparatively small forecast errors, the model can identify spatially coherent and meteorologically relevant patterns in climate data. These patterns often align with known atmospheric dynamics, such as cyclone structures, frontal systems, and moisture gradients, thereby supporting the physical interpretability of the model in favourable situations. Feature importance analysis further quantifies the influence of key meteorological variables like sea level pressure and 2 m temperature on forecasts. It also highlights the model's ability to use long-term temporal dependencies, suggesting potential for capturing extended climate signals in subseasonal forecasting. At the same time, in poorly predicted and extreme cases, the XAI diagnostics reveal more diffuse attention patterns and underdispersion, underlining the need for further model development.

These findings have important implications for subseasonal weather forecasting in Europe and AI applications in climate science. This study demonstrates that integrating XAI techniques into deep learning models enhances scientific transparency and provides useful diagnostic insights into both strengths and limitations of the ConvLSTM2D configuration, such as insufficient forecast dispersion and challenges in predicting extreme weather. These insights can guide improvements in model architecture, training strategies, and feature selection. Future research should build on these findings to develop more advanced models; for example, incorporating established climate indices and teleconnection patterns, enlarging the training sample, and exploring alternative loss functions could further improve forecast accuracy and robustness.

CRedit authorship contribution statement

Jing Chen: Data curation, Methodology, Software, Writing – original draft. **Ali Mohammed Mansoor Alsahag:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Seyed Sahand Mohammadi Ziabari:** Conceptualization, Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Supplementary exploratory figures

See Figs. 13–16.

Data availability

Data will be made available on request.

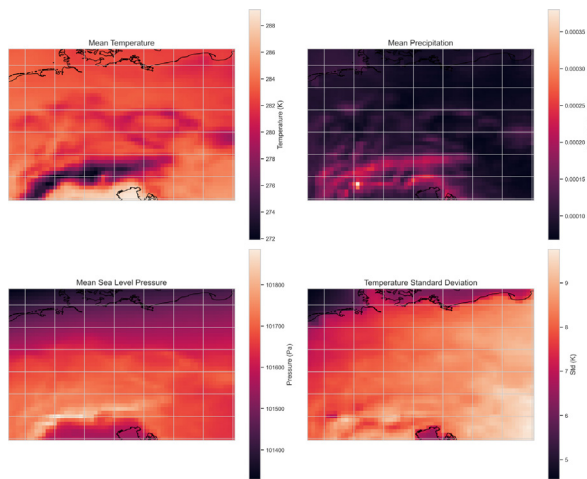


Fig. 15. Mean temperature. This spatial map displays the average temperature distribution across the region of interest, calculated from the ERA5 dataset over the considered time period.

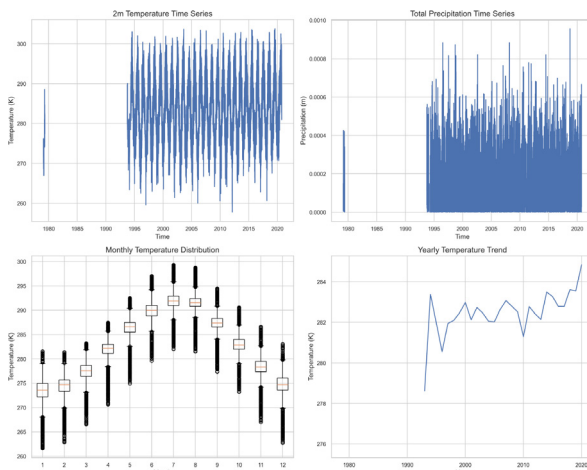


Fig. 16. Temperature time series. This time series plot shows the variation of 2 m temperature over the years, providing a temporal overview of temperature changes in the ERA5 dataset.

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